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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025

IS4224-1 – SOFTWARE DEFINED NETWORKS
(For ISE)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five** full questions choosing **Two** full questions from **Unit – I & Unit – II** each and **One** full question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. In traditional networks, the control plane is located in:
A) Centralized servers
B) Distributed across devices
C) Cloud platforms
D) Only in routers
2. Which year is considered the formal beginning of SDN with OpenFlow's introduction?
A) 2002
B) 2005
C) 2008
D) 2012
3. SDN is primarily designed to provide:
A) Hardware-based routing
B) Programmability and flexibility
C) Faster fiber cables
D) IPv6 only networks
4. Which of the following is NOT a motivation for SDN?
A) Simplified management
B) Vendor independence
C) Higher operational costs
D) Faster innovation
5. Which layer does the SDN controller logically sit in?
A) Application layer
B) Control layer
C) Infrastructure layer
D) Physical layer
6. In traditional networking, the control plane is:
A) Centralized
B) Distributed
C) Cloud-based
D) Virtualized
7. The data plane in SDN is responsible for:
A) Making routing decisions
B) Forwarding packets
C) Enforcing policies
D) Writing applications
8. Which of the following is a benefit of centralized control in SDN?
A) Vendor lock-in
B) Global network view
C) Increased complexity
D) Higher latency
9. What is the purpose of Mininet in SDN?
A) Network simulation
B) Hardware manufacturing
C) Traffic encryption
D) Data storage
10. Which SDN application is used for network slicing in 5G networks?
A) Plexxi
B) Cisco OnePK
C) Network Slicing
D) OpenFlow
11. What is the primary function of the control plane in SDN?
A) Forwarding data packets
B) Managing network devices
C) Making routing decisions
D) Encrypting traffic
12. Which SDN application is used for network monitoring and analytics?
A) Big Switch Networks
B) Floodlight
C) Ryu
D) POX



13. Which SDN application is used for software-defined storage?
 A) NetApp Development B) OpenFlow
 C) Ryu D) POX
14. Which benefit is common to all SDN environments?
 A) Reduced flexibility B) Centralized management
 C) Hardware dependency D) Manual configuration
15. An SDN use case in cloud service providers includes:
 A) Automated VM network provisioning B) Static VLAN management
 C) Manual IP addressing D) Fixed routing tables
16. A key SDN application in enterprises is:
 A) Dynamic load balancing B) Static routing
 C) Manual firewall configuration D) Hardware-only traffic management
17. Which SDN technology allows network operators to define traffic priorities?
 A) OpenFlow B) VLANs
 C) Static routing D) Manual packet forwarding
18. SDN open-source platforms promote:
 A) Vendor lock-in B) Interoperability and innovation
 C) Hardware dependency D) Manual configuration
19. An SDN use case in service provider networks includes:
 A) Static IP assignment B) Dynamic service chaining
 C) Manual hardware updates D) Fixed routing tables
20. SDN in data centers can improve:
 A) Manual device configuration B) Network agility and resource utilization
 C) Static routing policies D) Hardware dependence

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the key benefits of separating the control and data planes in a network element, and describe how this separation improves network functionality.	8	L*2	1	1
	b) Demonstrate how to create an MPLS overlay network over an OSPF underlay, using appropriate configuration steps or diagrams to illustrate the process.	8	L3	1	2
2.	a) Explain the concept of a distributed control plane in networking, and describe how the principle of eventual consistency applies within this model.	8	L2	1	1
	b) Apply your understanding of network orchestration to illustrate its layout and operational workflow using a diagram, and analyze the limitations associated with orchestration management.	8	L3	1	2
3.	a) Explain how technological advancements are influencing the evolving requirements of data centers, and describe the emerging needs in terms of infrastructure, scalability, and management.	8	L2	1	1
	b) Explain the architecture of a LAN Emulation (LANE) network, and describe the functions of the role-specific servers involved in its operation.	8	L2	1	1
Unit – II					
4.	a) Apply your understanding of OpenFlow v1.0 to explain the basic functions of an OpenFlow switch and its interaction with the controller. Illustrate the switch architecture and its relationship with the controller using a neat diagram.	8	L2	2	1

b)	Explain the standard and reserved virtual port types defined in an OpenFlow v1.1 switch, and describe their significance in packet forwarding and control operations.	8	L2	2	1
5. a)	Explain how OpenFlow v1.3 enables parallelism in communication between the switch and controller using auxiliary connections, and describe their significance in improving control channel efficiency.	8	L2	2	1
b)	Apply your understanding of OpenFlow v1.0 to illustrate the basic packet forwarding mechanism in a switch. Use a suitable diagram to demonstrate how packets are matched and forwarded based on flow table entries.	8	L3	2	2
6. a)	Explain the role of an SDN controller in a software-defined network, and describe its core modules and interfaces with the help of an anatomy diagram.	8	L2	2	1
b)	Explain the role of SDN applications in the Software-Defined Networking paradigm, and describe their key responsibilities in managing and optimizing network behavior.	8	L2	2	1
Unit – III					
7. a)	Explain the key characteristics of Stateless Transport Tunneling (STT), and describe its packet format used for tunneling in data center networks.	8	L2	3	1
b)	Apply your understanding of Software-Defined Networking (SDN) to explain how SDN overlays and Open SDN approaches help overcome limitations related to MAC address table size and the maximum number of VLANs. Illustrate the solution with a suitable diagram.	8	L3	3	2
8. a)	Explain the key networking demands emerging in modern data center environments due to technological advancements, and describe how these demands influence data center design and operations.	8	L2	3	1
b)	Explain the MAC-in-MAC (IEEE 802.1ah) and Q-in-Q (IEEE 802.1ad) encapsulation methods used in the Shortest Path Bridging (SPB) standard, and describe their roles in enhancing scalability and isolation in Ethernet networks.	8	L2	3	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
